Attention Based Spatial-Temporal Graph Convolutional Networks for Traffic Flow Forecasting

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Abstract

Forecasting the traffic flows is a critical issue for researchers and practitioners in the field of transportation. However, it is very challenging since the traffic flows usually show high nonlinearities and complex patterns. Most existing traffic flow prediction methods, lacking abilities of modeling the dynamic spatial-temporal correlations of traffic data, thus cannot yield satisfactory prediction results. In this paper, we propose a novel attention based spatial-temporal graph convolutional network (ASTGCN) model to solve traffic flow forecasting problem. ASTGCN mainly consists of three independent components to respectively model three temporal properties of traffic flows, i.e., recent, daily-periodic and weekly-periodic dependencies. More specifically, each component contains two major parts: 1) the spatial-temporal attention mechanism to effectively capture the dynamic spatial-temporal correlations in traffic data; 2) the spatial-temporal convolution which simultaneously employs graph convolutions to capture the spatial patterns and common standard convolutions to describe the temporal features. The output of the three components are weighted fused to generate the final prediction results. Experiments on two real-world datasets from the Caltrans Performance Measurement System (PeMS) demonstrate that the proposed ASTGCN model outperforms the state-of-the-art baselines.

Introduction

Recently, many countries are committed to vigorously develop the Intelligent Transportation System (ITS) (Zhang et al. 2011) to help for efficient traffic management. Traffic forecasting is an indispensable part of ITS, especially on the highway which has large traffic flows and fast driving speed. Since the highway is relatively closed, once a congestion occurs, it will seriously affect the traffic capacity. Traffic flow is a fundamental measurement reflecting the state of the highway. If it can be predicted accurately in advance, according to this, traffic management authorities will be able to guide vehicles more reasonably to enhance the running efficiency of the highway network.

Highway traffic flow forecasting is a typical problem of spatial-temporal data forecasting. Traffic data are recorded at fixed points in time and at fixed locations distributed

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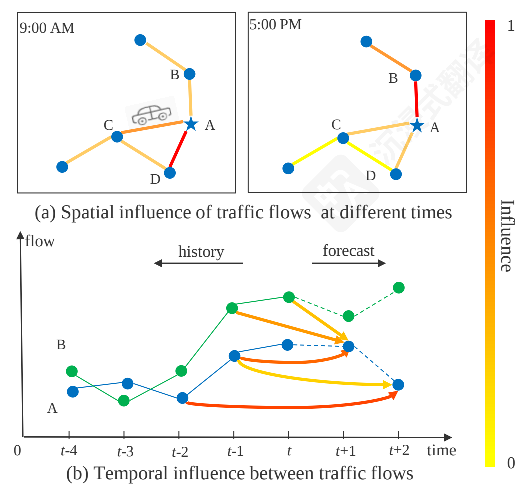

Figure 1: The spatial-temporal correlation diagram of traffic flow

in continuous space. Apparently, the observations made at neighboring locations and time stamps are not independent but dynamically correlated with each other. Therefore, the key to solve such problems is to effectively extracting the spatial-temporal correlations of data. Fig. 1 demonstrates the spatial-temporal correlations of traffic flows (also can be vehicle speed, lane occupancy, etc.). The bold line between two points represents their mutual influence strength. The darker the color of line is, the greater the influence is. In the spatial dimension (Fig. 1(a)), we can find that different locations have different impacts on A and even a same location has varying influence on A as time goes by. In the temporal dimension (Fig. 1(b)), the historical observations of different locations have varying impacts on A's traffic states at different times in the future. In conclusion, the correlations in traffic data on the highway network show strong dynamics in both the spatial dimension and temporal dimension. How to explore nonlinear and complex spatial-temporal data to discover its inherent spatial-temporal patterns and to make accurate traffic flow predictions is a very challenging issue.

Fortunately, with the development of the transportation industry, many cameras, sensors and other information collection devices have been deployed on the highway. Each

基于注意力机制的时空图卷积网络用于交通流量预测

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摘要

预测交通流量是交通领域研究人员和从业者的关键问题。然而，这非常具有挑战性，因为交通流量通常表现出高度的非线性和复杂的模式。大多数现有的交通流量预测方法缺乏建模交通数据动态时空相关性的能力，因此无法获得令人满意的预测结果。在本文中，我们提出了一种基于注意力机制的时空图卷积网络（ASTGCN）模型来解决交通流量预测问题。ASTGCN主要由三个独立组件组成，分别用于建模交通流的三种时间属性，即近期、日周期和周周期依赖性。更具体地说，每个组件包含两个主要部分：1）时空注意力机制，用于有效捕获交通数据中的动态时空相关性；2）时空卷积，同时采用图卷积来捕获空间模式，并使用标准卷积来描述时间特征。三个组件的输出被加权融合以生成最终预测结果。在来自加州交通性能测量系统（PeMS）的两个真实世界数据集上的实验表明，所提出的ASTGCN模型优于现有最佳基线。

引言

近年来，许多国家致力于大力发展智能交通系统（ITS）（Zhang 等人，2011 年），以帮助实现高效的交通管理。交通预测是智能交通系统不可或缺的一部分，尤其是在高速公路上，其交通流量大且行驶速度快。由于高速公路相对封闭，一旦发生拥堵，将严重影响交通能力。交通流量是反映高速公路状态的基本指标。如果能够提前准确预测，根据这一结果，交通管理部门将能够更合理地引导车辆，以提高公路网络的运行效率。

高速公路交通流量预测是一个典型的时空数据预测问题。交通数据在固定的时间点和固定位置上被记录，并分布

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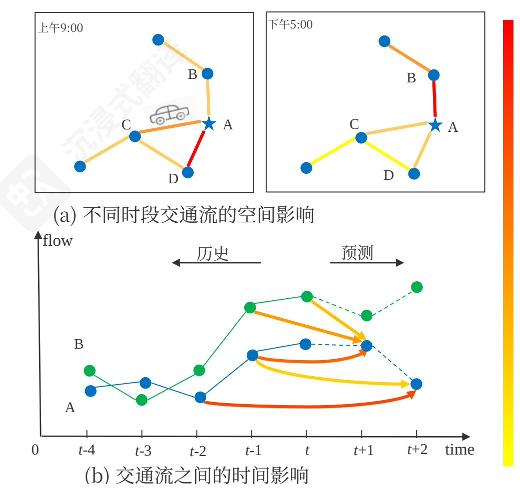


图 1：交通流时空相关图

在连续空间中。显然，在相邻位置和时间戳上的观测值不是独立的，而是相互动态关联的。因此，解决这类问题的关键在于有效提取数据的时空相关性。图1展示了交通流量的时空相关性（也可以是车辆速度、车道占用率等）。两点之间的粗线表示它们相互影响强度。线的颜色越深，影响越大。在空间维度（图1(a)）中，我们可以发现不同位置对A的影响不同，甚至同一位置随时间推移对A的影响也在变化。在时间维度（图1(b)）中，不同位置的历史观测值对未来不同时间点A的交通状态具有不同影响。总之，高速公路网络上的交通数据在空间维度和时间维度上均表现出强烈的动态性。如何探索非线性、复杂的时空数据，发现其内在的时空模式，并做出准确的交通流量预测，是一个非常具有挑战性的问题。

幸运的是，随着交通业的发展，许多摄像头、传感器和其他信息采集设备已被部署在高速公路上。每

device is placed at a unique geospatial location, constantly generating time series data about traffic. These devices have accumulated a large amount of rich traffic time series data with geographic information, providing a solid data foundation for traffic forecasting. Many researchers have already made great efforts to solve such problems. Early, time series analysis models are employed for traffic prediction problems. Yet, it is difficult for them to handle the unstable and nonlinear data in practice. Later, traditional machine learning methods are developed to model more complex data, but it is still difficult for them to simultaneously consider the spatial-temporal correlations of high-dimensional traffic data. Moreover, the prediction performances of this kind of methods rely heavily on feature engineering, which often requires lots of experiences from experts in the corresponding domain. In recent years, many researchers use deep learning methods to deal with high-dimensional spatial-temporal data, i.e., convolutional neural networks (CNN) are adopted to effectively extract the spatial features of grid-based data; graph convolutional neural networks (GCN) are used for describing spatial correlation of graph-based data. However, these methods still fail to simultaneously model the spatial-temporal features and dynamic correlations of traffic data.

In order to tackle the above challenges, we propose a novel deep learning model: ***A**ttention based **S**patial-**T**emporal **G**raph **C**onvolution **N**etwork* (ASTGCN) to collectively predict traffic flow at every location on the traffic network. This model can process the traffic data directly on the original graph-based traffic network and effectively capture the dynamic spatial-temporal features. The main contributions of this paper are summarized as follows:

- We develop a spatial-temporal attention mechanism to learn the dynamic spatial-temporal correlations of traffic data. Specifically, a spatial attention is applied to model the complex spatial correlations between different locations. A temporal attention is applied to capture the dynamic temporal correlations between different times.
- A novel spatial-temporal convolution module is designed for modeling spatial-temporal dependencies of traffic data. It consists of graph convolutions for capturing spatial features from the original graph-based traffic network structure and convolutions in the temporal dimension for describing dependencies from nearby time slices.
- Extensive experiments are carried out on real-world highway traffic datasets, which verify that our model achieves the best prediction performances compared to the existing baselines.

Related work

Traffic forecasting After years of continuous researches and practices, many achievements have been made in the studies about traffic forecasting. The statistical models used for traffic prediction include HA, ARIMA (Williams and Hoel 2003), VAR (Zivot and Wang 2006), etc. These approaches require data to satisfy some assumptions, but traffic data is too complex to satisfy these assumptions, so they usually perform poorly in practice. Machine learning methods such as KNN (Van Lint and Van Hinsbergen 2012) and SVM (Jeong et al. 2013) can model more complex data, but they need careful feature engineering. Since deep learning has brought about breakthroughs in many domains, such as speech recognition and image processing, more and more researchers apply deep learning to spatial-temporal data prediction. Zhang et al. (2018) designed a ST-ResNet model based on the residual convolution unit to predict crowd flows. Yao et al. (2018b) proposed a method to predict traffic by integrating CNN and long-short term memory (LSTM) to jointly model both spatial and temporal dependencies. Yao et al. (2018a) further proposed a Spatial-Temporal Dynamic Network for taxi demand prediction which can learn the similarity between locations dynamically. Although the spatial-temporal features of the traffic data can be extracted by these model, their limitation is that the input must be standard 2D or 3D grid data.

Convolutions on graphs The traditional convolution can effectively extract the local patterns of data, but it can only be applied for the standard grid data. Recently, the graph convolution generalizes the traditional convolution to data of graph structures. Two mainstreams of graph convolution methods are the spatial methods and the spectral methods. The spatial methods directly perform convolution filters on a graph's nodes and their neighbors. So, the core of this kind of methods is to select the neighborhood of nodes. Niepert, Ahmed, and Kutzkov (2016) proposed a heuristic linear method to select the neighborhood of every center node, which achieved good results in social network tasks. Li et al. (2018) introduced graph convolutions into human action recognition tasks. Several partitioning strategies were proposed here to divide the neighborhood of each node into different subsets and to ensure the numbers of each node's subsets are equal. The spectral methods, in which the locality of the graph convolution is considered by spectral analysis. A general graph convolution framework based on the Graph Laplacian is proposed by Bruna et al. (2014), then Defferrard, Bresson, and Vandergheynst (2016) optimized the method by using Chebyshev polynomial approximation to realize eigenvalue decomposition. Yu, Yin, and Zhu (2018) proposed a gated graph convolution network for traffic prediction based on this method, but the model does not consider the dynamic spatial-temporal correlations of traffic data.

Attention mechanism Recently, attention mechanisms have been widely used in various tasks such as natural language processing, image caption and speech recognition. The goal of the attention mechanism is to select information that is relatively critical to the current task from all input. Xu et al. (2015) proposed two attention mechanisms in the image description task and adopted a visualization method to intuitively show the effect of the attention mechanism. For classifying nodes of a graph, Velickovic et al. (2018) leveraged self-attentional layers to process graph-structured data by neural networks and achieved state-of-the-art results. To forecast the time series, Liang et al. (2018) proposed a multi-level attention network to adaptively adjust the correlations among multiple geographic sensor time series. However, it is time-consuming in practice since a separate model needs

个设备都放置在独特的地理空间位置，不断生成关于交通的时间序列数据。这些设备积累了大量包含地理信息丰富的时间序列交通数据，为交通预测提供了坚实的基础。许多研究人员已经为解决此类问题做出了巨大努力。早期，时间序列分析模型被用于交通预测问题。然而，它们在实践中难以处理不稳定和非线性数据。后来，传统机器学习方法被开发出来以建模更复杂的数据，但它们仍然难以同时考虑高维交通数据的时空相关性。此外，这类方法的预测性能严重依赖于特征工程，这通常需要相应领域专家大量的经验。近年来，许多研究人员使用深度学习方法处理高维时空数据，即采用卷积神经网络（CNN）有效提取基于网格数据的空间特征；使用图卷积神经网络（GCN）描述基于图数据的空间相关性。然而，这些方法仍然无法同时建模交通数据的时空特征和动态相关性。

为了应对上述挑战，我们提出了一种新的深度学习模型：基于注意力的时空图卷积网络-(ASTGCN)，用于联合预测交通网络中每个位置的交通流量。该模型可以直接在原始的基于图的交通网络上处理交通数据，并有效捕捉动态时空特征。本文的主要贡献总结如下：

- 我们开发了一种时空注意力机制，用于学习交通数据的动态时空相关性。具体来说，空间注意力用于建模不同位置之间的复杂空间相关性。时间注意力用于捕捉不同时间之间的动态时间相关性。
- 我们设计了一种新的时空卷积模块，用于建模交通数据的时空依赖关系。它由图卷积组成，用于从原始的基于图的交通网络结构中捕获空间特征，以及时间维度上的卷积，用于描述来自邻近时间片的依赖关系。
- 我们在真实世界的高速交通数据集上进行了大量实验，验证了我们的模型与现有基线相比实现了最佳预测性能。

相关工作

交通预测 经过多年的持续研究和实践，交通预测研究已取得许多成果。用于交通预测的统计模型包括HA、ARIMA（Williams和Hoel 2003年）、VAR（Zivot和Wang 2006年）等。这些方法需要数据满足某些假设，但交通数据过于复杂，难以满足这些假设，因此在实践中通常表现不佳。KNN（Van Lint和Van Hinsbergen 2012年）和SVM（Jeong等人 2013年）等机器学习方法可以建模更复杂的数据，但它们需要仔细的特征工程。

由于深度学习在语音识别和图像处理等多个领域取得了突破，越来越多的研究人员将其应用于时空数据预测。Zhang等人（2018年）设计了一种基于残差卷积单元的ST-ResNet模型来预测人群流量。Yao等人（2018b年）提出了一种通过整合CNN和长短期记忆（LSTM）来预测交通的方法，以联合建模空间和时间依赖性。Yao等人（2018a年）进一步提出了一种用于出租车需求预测的时空动态网络，该网络可以动态学习位置之间的相似性。尽管这些模型可以提取交通数据的时空特征，但它们的局限性在于输入必须是标准的2D或3D网格数据。

图卷积 传统卷积可以有效地提取数据的局部模式，但它只能应用于标准网格数据。最近，图卷积将传统卷积推广到图结构数据。图卷积方法主要有空间方法和谱方法两种。空间方法直接在图的节点及其邻居上执行卷积滤波器。因此，这类方法的核心是选择节点的邻域。Niepert、Ahmed和Kutzkov（2016）提出了一种启发式线性方法来选择每个中心节点的邻域，在社交网络任务中取得了良好结果。Li等人（2018）将图卷积引入人类动作识别任务。这里提出了几种划分策略，将每个节点的邻域划分为不同的子集，并确保每个节点的子集数量相等。谱方法通过谱分析考虑图卷积的局部性。Bruna等人（2014）提出了基于图拉普拉斯的通用图卷积框架，然后Defferrard、Bresson和Vandergheynst（2016）

通过使用切比雪夫多项式逼近来改进该方法，以实现特征值分解。Yu、Yin和Zhu（2018年）提出了基于此方法的门控图卷积网络用于交通预测，但该模型未考虑交通数据的动态时空相关性。

注意力机制 近年来，注意力机制已被广泛应用于自然语言处理、图像描述和语音识别等各种任务中。注意力机制的目标是从所有输入中选取相对当前任务较为关键的信息。徐等人（2015）在图像描述任务中提出了两种注意力机制，并采用可视化方法直观地展示了注意力机制的效果。对于图节点的分类，Velickovic等人（2018）利用自注意力层通过神经网络处理图结构数据，并取得了当前最佳结果。为了预测时间序列，梁等人（2018）提出了一种多层注意力网络，以自适应地调整多个地理传感器时间序列之间的相关性。然而，在实际应用中耗时较长，因为需要对每个时间序列训练一个单独的模型。

to be trained for each time series.

Motivated by the studies mentioned above, considering the graph structure of the traffic network and the dynamic spatio-temporal patterns of the traffic data, we simultaneously employ graph convolutions and the attention mechanisms to model the network-structure traffic data.

Preliminaries

Traffic Networks

In this study, we define a traffic network as an undirected graph $G = (V, E, \mathbf{A})$, as shown in Fig. 2(a), where V is a finite set of $|V| = N$ nodes; E is a set of edges, indicating the connectivity between the nodes; $\mathbf{A} \in \mathbb{R}^{N\times N}$ denotes the adjacency matrix of graph G . Each node on the traffic network G detects F measurements with the same sampling frequency, that is, each node generates a feature vector of length F at each time slice, as shown by the solid lines in Fig. 2(b).

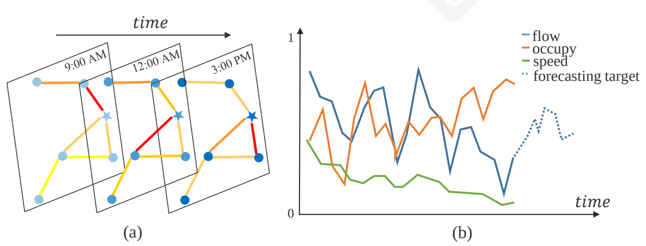


Figure 2: (a) The spatial-temporal structure of traffic data, where the data at each time slice forms a graph; (b) Three measurements are detected on a node and the future traffic flow is the forecasting target. Here, all measurements are normalized to [0,1].

Traffic Flow Forecasting

Suppose the f-th time series recorded on each node in the traffic network G is the traffic flow sequence, and $f \in (1, ..., F)$. We use $x_t^{c,i} \in \mathbb{R}$ to denote the value of the c-th feature of node i at time t , and $\mathbf{x}_t^i \in \mathbb{R}^F$ denotes the values of all the features of node i at time t. $\mathbf{X}_t = (\mathbf{x}_t^1, \mathbf{x}_t^2, ..., \mathbf{x}_t^N)^T \in \mathbb{R}^{N\times F}$ denotes the values of all the features of all the nodes at time t. $\boldsymbol{\mathcal{X}} = (\mathbf{X}_1, \mathbf{X}_2, ..., \mathbf{X}_\tau)^T \in \mathbb{R}^{N\times F\times\tau}$ denotes the value of all the features of all the nodes over τ time slices. In addition, we set $y_t^i = x_t^{f,i} \in \mathbb{R}$ to represent the traffic flow of node i at time t in the future.

Problem. Given $\boldsymbol{\mathcal{X}}$, all kinds of the historical measurements of all the nodes on the traffic network over past τ time slices, predict future traffic flow sequences $\mathbf{Y} = (\mathbf{y}^1, \mathbf{y}^2, ..., \mathbf{y}^N)^T \in \mathbb{R}^{N\times T_p}$ of all the nodes on the whole traffic network over the next T_p time slices, where $\mathbf{y}^i = (y_{\tau+1}^i, y_{\tau+2}^i, ..., y_{\tau+T_p}^i) \in \mathbb{R}^{T_p}$ denotes the future traffic flow of node i from $\tau + 1$.

Attention Based Spatial-Temporal Graph Convolutional Networks

Fig. 3 presents the overall framework of the ASTGCN model proposed in this paper. It consists of three independent components with the same structure, which are designed to respectively model the recent, daily-periodic and weekly-periodic dependencies of the historical data.

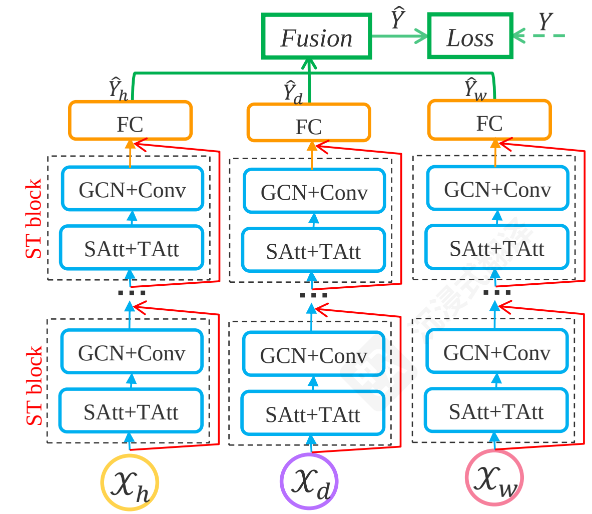


Figure 3: The framework of ASTGCN. SAtt: Spatial Attention; TAtt: Temporal Attention GCN: Graph Convolution; Conv: Convolution; FC: Fully-connected; ST block: Spatial-Temporal block.

Suppose the sampling frequency is q times per day. Assume that the current time is t_0 and the size of predicting window is T_p. As shown in Fig. 4, we intercept three time series segments of length T_h, T_d and T_w along the time axis as the input of the recent, daily-period and weekly-period component respectively, where T_h, T_d and T_w are all integer multiples of T_p. Details about the three time series segments are as follows:

(1) The *recent* segment:
$\boldsymbol{\mathcal{X}}_h = (\mathbf{X}_{t_0-T_h+1}, \mathbf{X}_{t_0-T_h+2}, ..., \mathbf{X}_{t_0}) \in \mathbb{R}^{N\times F\times T_h}$, a segment of historical time series directly adjacent to the predicting period, as shown by the green part of Fig. 4. Intuitively, the formation and dispersion of traffic congestions are gradual. So, the just past traffic flows inevitably have influence on the future traffic flows.

(2) The *daily-periodic* segment:
$\boldsymbol{\mathcal{X}}_d = (\mathbf{X}_{t_0-(T_d/T_p)*q+1}, ..., \mathbf{X}_{t_0-(T_d/T_p)*q+T_p}, \mathbf{X}_{t_0-(T_d/T_p-1)*q+1}, ..., \mathbf{X}_{t_0-(T_d/T_p-1)*q+T_p}, ..., \mathbf{X}_{t_0-q+1}, ..., \mathbf{X}_{t_0-q+T_p}) \in \mathbb{R}^{N\times F\times T_d}$ consists of the segments on the past few days at the same time period as the predicting period, as shown by the red part of Fig. 4. Due to the regular daily routine of people, traffic data may show repeated patterns, such as the daily morning peaks. The purpose of the daily-period component is to model the daily periodicity of traffic data.

(3) The *weekly-periodic* segment:
$\boldsymbol{\mathcal{X}}_w = (\mathbf{X}_{t_0-7*(T_w/T_p)*q+1}, ..., \mathbf{X}_{t_0-7*(T_w/T_p)*q+T_p},$

需要为每个时间序列单独训练模型。

受上述研究启发，考虑交通网络的图结构以及交通数据的动态时空模式，我们同时采用图卷积和注意力机制来建模网络结构交通数据。

预备知识

交通网络

在本研究中，我们将交通网络定义为一个无向图 $G = (V, E, \mathbf{A})$，如图 2(a) 所示，其中 V 是一个有限节点集； E 是一个表示节点之间连接性的边集 $\mathbf{A} \in \mathbb{R}^{N\times N}$； G 表示图 G 的邻接矩阵。交通网络 G 上的每个节点以相同的采样频率检测 F 个测量值，也就是说，每个节点在每个时间片上生成一个长度为 F 的特征向量，如图 2(b) 中的实线所示。

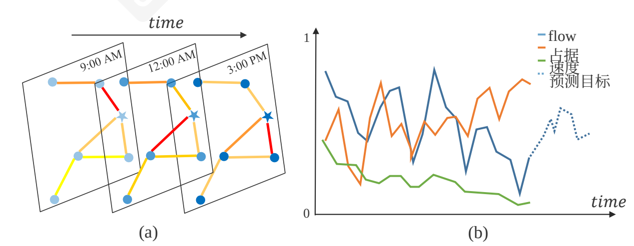


图 2：(a) 交通数据的时空结构，其中每个时间片上的数据形成一个图；(b) 一个节点上检测到三个测量值，未来的交通流量是预测目标。这里，所有测量值都归一化到 [0,1]。

交通流量预测

假设交通网络 G 上的每个节点记录的 f-th 时间序列是交通流量序列，f $\in (1, ..., F)$。我们用 $x_t^{c\ i} \in \mathbb{R}$ 表示该值的

c-th f特征 o f node i at ti某 t，一个 d $\mathbf{x}_t^i \in \mathbb{R}^F$ d

该注解标明了节点 i 在时间 t 的所有特征值。$\mathbf{X}_t = (\mathbf{x}_t^1, \mathbf{x}_t^2, ..., \mathbf{x}_t^N)^T \in \mathbb{R}^{N\times F}$ 表示所有节点在时间 t 的所有特征值。$\boldsymbol{\mathcal{X}} = (\mathbf{X}_1, \mathbf{X}_2, ..., \mathbf{X}_\tau)^T \in \mathbb{R}^{N\times F\times\tau}$ 表示所有节点在 τ 时间切片内的所有特征值。此外，我们设置 $y_t^i = x_t^{f\ i} \in \mathbb{R}$ 来表示节点 i 在时间 t 的未来流量。

问题。给定X，所有节点在交通网络上的历史测量数据，跨越过去的τ时间片，预测未来整个交通网络所有节点的交通流序列 $\mathbf{Y} = (\mathbf{y}^1, \mathbf{y}^2, ..., \mathbf{y}^N)^T \in \mathbb{R}^{N\times T_p}$，其中 $\mathbf{y}^i = (y_{\tau+1}^i, y_{\tau+2}^i, ..., y_{\tau+T_p}^i) \in \mathbb{R}^{T_p}$ 表示节点 i 从 $\tau+1$ 的未来交通流。

基于注意力的时空图卷积网络

图3展示了本文提出的ASTGCN模型的整体框架。它由三个具有相同结构的独立组件组成，分别用于建模历史数据的近期、日周期和周周期依赖关系。

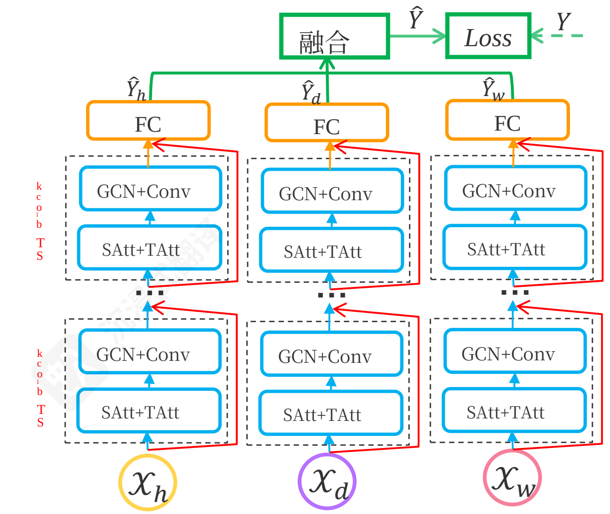


图 3：ASTGCN 的框架。SAtt：空间注意力；TAtt：时间注意力 GCN：图卷积；Conv：卷积；FC：全连接；STblock：时空块。

假设采样频率为每天 q 次。假设当前时间为 t_0，预测窗口大小为 T_p。如图 4 所示，我们沿时间轴截取长度为 T_h、T_d 和 T_w 的三个时间序列片段作为近期、日周期和周周期的输入，其中 T_h、T_d 和 T_w 均为 T_p 的整数倍。三个时间序列片段的详细信息如下：

(1) 最近一段：
$\boldsymbol{\mathcal{X}}_h = (\mathbf{X}_{t_0-T_h+1}, \mathbf{X}_{t_0-T_h+2}, ..., \mathbf{X}_{t_0}) \in \mathbb{R}^{N\times F\times T_h}$，即历史时间序列中与预测期直接相邻的一段，如图4的绿色部分所示。直观地看，交通拥堵的形成和消散是逐渐发生的。因此，刚刚过去的交通流量不可避免地对未来的交通流量产生影响。

(2) 日周期段：(
$\boldsymbol{\mathcal{X}}_d = \mathbf{X}_{t_0-(T_d/T_p)*q+1}, ..., \mathbf{X}_{t_0-(T_d/T_p)*q+T_p}, \mathbf{X}_{t_0-(T_d/T_p-1)*q+1}, ..., \mathbf{X}_{t_0-(T_d/T_p-1)*q+T_p}, ..., \mathbf{X}_{t_0-q+1} ... \mathbf{X}_{t_0-q+T_p}) \in \mathbb{R}^{N\times F\times T_d}$，，由过去几天与预测周期相同时间段内的片段组成，如图4中的红色部分所示。由于人们的日常生活规律，交通数据可能会出现重复模式，例如每日早高峰。每日周期成分的目的是对交通数据的每日周期性进行建模。

(3) 周周期段：(
$\boldsymbol{\mathcal{X}}_w = \mathbf{X}_{t_0-7*(T_w/T_p)*q+1}, ..., \mathbf{X}_{t_0-7*(T_w/T_p)*q+T_p},$

$\mathbf{X}_{t_0-7*(T_w/T_p-1)*q+1}, \cdots, \mathbf{X}_{t_0-7*(T_w/T_p-1)*q+T_p}, \cdots,$ $\mathbf{X}_{t_0-7*q+1}, \cdots, \mathbf{X}_{t_0-7*q+T_p}) \in \mathbb{R}^{F\times N\times T_w}$ is composed of the segments on last few weeks, which have the same week attributes and time intervals as the forecasting period, as shown by the blue part of Fig. 4. Usually, the traffic patterns on Monday have a certain similarity with the traffic patterns on Mondays in history, but may be greatly different from those on weekends. Thus, the weekly-period component is designed to capture the weekly periodic features in traffic data.

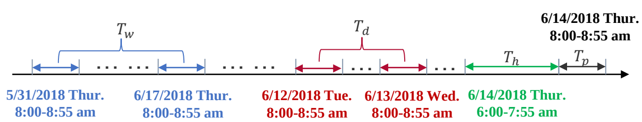


Figure 4: An example of constructing the input of time series segments (suppose the size of predicting window is 1 hour, and T_h, T_d and T_w are twice of T_p.

The three components share the same network structure and each of them consists of several spatial-temporal blocks and a fully-connected layer. There are a spatial-temporal attention module and a spatial-temporal convolution module in each spatial-temporal block. In order to optimize the training efficiency, we adopted a residual learning framework (He et al. 2016) in each component. In the end, the outputs of the three components are further merged based on a parameter matrix to obtain the final prediction result. The overall network structure is elaborately designed to describe the dynamic spatial-temporal correlations of traffic flows.

Spatial-Temporal Attention

A novel spatial-temporal attention mechanism is proposed in our model to capture the dynamic spatial and temporal correlations on the traffic network (as described in Fig. 1). It contains two kinds of attentions, i.e., spatial attention and temporal attention.

Spatial attention In the spatial dimension, the traffic conditions of different locations have influence among each other and the mutual influence is highly dynamic. Here, we use an attention mechanism (Feng et al. 2017) to adaptively capture the dynamic correlations between nodes in the spatial dimension.

Take the spatial attention in the recent component as an example:

$$\mathbf{S} = \mathbf{V}_s \cdot \sigma((\boldsymbol{\mathcal{X}}_h^{(r-1)}\mathbf{W}_1)\mathbf{W}_2(\mathbf{W}_3\boldsymbol{\mathcal{X}}_h^{(r-1)})^T + \mathbf{b}_s) \quad (1)$$

$$\mathbf{S}'_{i,j} = \frac{\exp(\mathbf{S}_{i,j})}{\sum_{j=1}^{N}\exp(\mathbf{S}_{i,j})} \quad (2)$$

where $\boldsymbol{\mathcal{X}}_h^{(r-1)} = (\mathbf{X}_1, \mathbf{X}_2, ...\mathbf{X}_{T_{r-1}}) \in \mathbb{R}^{N\times C_{r-1}\times T_{r-1}}$ is the input of the r^{th} spatial-temporal block. C_{r-1} is the number of channels of the input data in the r^{th} layer. When $r = 1$, $C_0 = F$. T_{r-1} is the length of the temporal dimension in the r^{th} layer. When $r = 1$, in the recent component $T_0 = T_h$ (in the daily-period component $T_0 = T_d$ and in the weekly-period component $T_0 = T_w$). $\mathbf{V}_s, \mathbf{b}_s \in \mathbb{R}^{N\times N}$, $\mathbf{W}_1 \in \mathbb{R}^{T_{r-1}}$, $\mathbf{W}_2 \in \mathbb{R}^{C_{r-1}\times T_{r-1}}$, $\mathbf{W}_3 \in \mathbb{R}^{C_{r-1}}$ are learnable parameters and sigmoid σ is used as the activation function. The attention matrix $\mathbf{S}$ is dynamically computed according to the current input of this layer. The value of an element $\mathbf{S}_{i,j}$ in $\mathbf{S}$ semantically represents the correlation strength between node i and node j. Then a softmax function is used to ensure the attention weights of a node sum to one. When performing the graph convolutions, we will accompany the adjacency matrix $\mathbf{A}$ with the spatial attention matrix $\mathbf{S}' \in \mathbb{R}^{N\times N}$ to dynamic adjust the impacting weights between nodes.

Temporal attention In the temporal dimension, there exist correlations between the traffic conditions in different time slices, and the correlations are also varying under different situations. Likewise, we use an attention mechanism to adaptively attach different importance to data:

$$\mathbf{E} = \mathbf{V}_e \cdot \sigma(((\boldsymbol{\mathcal{X}}_h^{(r-1)})^T\mathbf{U}_1)\mathbf{U}_2(\mathbf{U}_3\boldsymbol{\mathcal{X}}_h^{(r-1)}) + \mathbf{b}_e) \quad (3)$$

$$\mathbf{E}'_{i,j} = \frac{\exp(\mathbf{E}_{i,j})}{\sum_{j=1}^{T_{r-1}}\exp(\mathbf{E}_{i,j})} \quad (4)$$

where $\mathbf{V}_e, \mathbf{b}_e \in \mathbb{R}^{T_{r-1}\times T_{r-1}}$, $\mathbf{U}_1 \in \mathbb{R}^N$, $\mathbf{U}_2 \in \mathbb{R}^{C_{r-1}\times N}$, $\mathbf{U}_3 \in \mathbb{R}^{C_{r-1}}$ are learnable parameters. The temporal correlation matrix $\mathbf{E}$ is determined by the varying inputs. The value of an element $\mathbf{E}_{i,j}$ in $\mathbf{E}$ semantically indicates the strength of dependencies between time i and j. At last, $\mathbf{E}$ is normalized by the softmax function. We directly apply the normalized temporal attention matrix to the input and get $\hat{\boldsymbol{\mathcal{X}}}_h^{(r-1)} = (\hat{\mathbf{X}}_1, \hat{\mathbf{X}}_2, ..., \hat{\mathbf{X}}_{T_{r-1}}) = (\mathbf{X}_1, \mathbf{X}_2, ..., \mathbf{X}_{T_{r-1}})\mathbf{E}' \in \mathbb{R}^{N\times C_{r-1}\times T_{r-1}}$ to dynamically adjust the input by merging relevant information.

Spatial-Temporal Convolution

The spatial-temporal attention module let the network automatically pay relatively more attention on valuable information. The input adjusted by the attention mechanism is fed into the spatial-temporal convolution module, whose structure is presented in Fig. 5. The spatial-temporal convolution module proposed here consists of a graph convolution in the spatial dimension, capturing spatial dependencies from neighborhood and a convolution along the temporal dimension, exploiting temporal dependencies from nearby times.

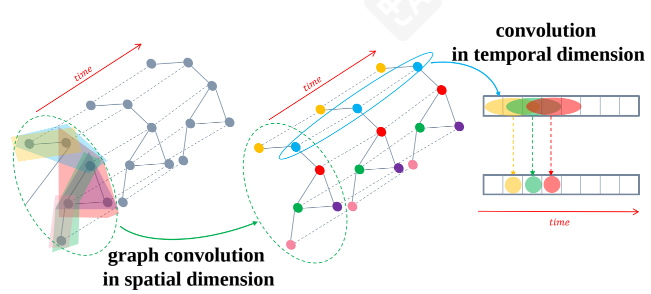


Figure 5: The architecture of spatial-temporal convolutions of ASTGCN.

$\mathbf{X}_{t_0-7*(T_w/T_p-1)*q+1}, \cdots, \mathbf{X}_{t_0-7*(T_w/T_p-1)*q+T_p}, \cdots,$ $\mathbf{X}_{t_0-7*q+1} \cdots \mathbf{X}_{t_0-7*q+T_p}) \in \mathbb{R}^{F\times N\times T_w}$，，由过去几周内具有相同周属性和时间间隔的片段组成，如图4中的蓝色部分所示。通常，周一的交通模式与历史上的周一交通模式有一定相似性，但可能与周末的交通模式差异很大。因此，周周期成分的设计目的是捕捉交通数据中的周周期特征。

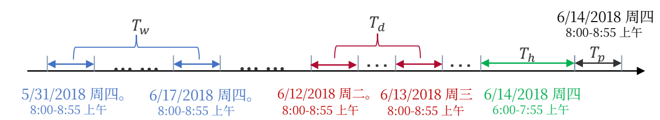


图4：构建时间序列片段输入的一个示例（假设预测窗口大小为1小时，且 T_h、T_d 和 T_w 是 T_p 的两倍）。

这三个组件共享相同的网络结构，并且每个组件都由多个时空块和一个全连接层组成。每个时空块中都有一个时空注意力模块和一个时空卷积模块。为了优化训练效率，我们在每个组件中采用了残差学习框架（He 等人，2016）。最后，三个组件的输出基于一个参数矩阵进一步合并，以获得最终预测结果。整个网络结构精心设计，用于描述交通流量的动态时空相关性。

时空注意力

我们的模型中提出了一种新的时空注意力机制，用于捕获交通网络上的动态时空相关性（如图1所示）。它包含两种注意力机制，即空间注意力和时间注意力。

空间注意力 在空间维度上，不同位置的交通状况相互影响，并且这种相互影响具有高度动态性。在这里，我们使用注意力机制（Feng 等人，2017）来自适应地捕获空间维度中节点之间的动态相关性。

以最近成分中的空间注意力为例：

$$\mathbf{S} = \mathbf{V}_s \cdot \sigma((\boldsymbol{\mathcal{X}}_h^{(r-1)}\mathbf{W}_1)\mathbf{W}_2(\mathbf{W}_3\boldsymbol{\mathcal{X}}_h^{(r-1)})^T + \mathbf{b}_s) \quad (1)$$

$$\mathbf{S}'_{i,j} = \frac{\exp(\mathbf{S}_{i,j})}{\sum_{j=1}^{N}\exp(\mathbf{S}_{i,j})} \quad (2)$$

在 $\boldsymbol{\mathcal{X}}_h^{(r-1)} = (\mathbf{X}_1, \mathbf{X}_2, ...\mathbf{X}_{T_{r-1}}) \in \mathbb{R}^{N\times C_{r-1}\times T_{r-1}}$ i 是输入的 r^{th} 时空块。C_{r-1} 是输入数据在 r^{th} 层的通道数。当 $r = 1$ $C_0 = F$. T_{r-1}，是 r^{th} 层中时间维度的长度。当 $r = 1$，在最近成分 $T_0 = T_h$（在日周期成分 $T_0 = T_d$ 中）每周周期组件 $T_0 = T_w$)。V_s、$\mathrm{b}_s \in \mathbb{R}^{N\times N}$、W $_1 \in \mathbb{R}^{T_{r-1}}$、$\mathrm{W}_2 \in \mathbb{R}^{C_{r-1}\times T_{r-1}}$、$\mathrm{W}_3 \in \mathbb{R}^{C_{r-1}}$ 是可学习的参数，sigmoid σ 用作激活函数。注意力矩阵 S 根据该层当前输入动态计算。S 中元素 $\mathbf{S}_{i\ j}$ 的值在语义上表示节点 i 和节点 j 之间的相关强度。然后使用 softmax 函数确保节点的注意力权重之和为 1。在进行图卷积时，我们将伴随邻接矩阵 A 和空间注意力矩阵 $\mathbf{S}' \in \mathbb{R}^{N\times N}$ 动态调整节点之间的影响权重。

时间注意力 在时间维度上，不同时间片之间的交通状况存在相关性，且这些相关性在不同情况下也会变化。同样地，我们使用注意力机制来自适应地为数据赋予不同的重要性：

$$\mathbf{E} = \mathbf{V}_e \cdot \sigma(((\boldsymbol{\mathcal{X}}_h^{(r-1)})^T\mathbf{U}_1)\mathbf{U}_2(\mathbf{U}_3\boldsymbol{\mathcal{X}}_h^{(r-1)}) + \mathbf{b}_e) \quad (3)$$

$$\mathbf{E}'_{i,j} = \frac{\exp(\mathbf{E}_{i,j})}{\sum_{j=1}^{T_{r-1}}\exp(\mathbf{E}_{i,j})} \quad (4)$$

其中 $\mathrm{V}_e, \mathrm{b}_e \in \mathbb{R}^{T_{r-1}\times T_{r-1}}$, $\mathrm{U}_1 \in \mathbb{R}^N$, $\mathrm{U}_2 \in \mathbb{R}^{C_{r-1}\times N}$, $\mathbf{U}_3 \in \mathbb{R}^{C_{r-1}}$ 是可学习参数。时间相关矩阵 E 由变化的输入决定。E 中元素 $\mathbf{E}_{i\ j}$ 的值在语义上表示时间 i 和 j 之间的依赖强度。最后，E 通过 softmax 函数进行归一化。我们直接将归一化的时间注意力矩阵应用于输入，得到 $\hat{\boldsymbol{\mathcal{X}}}_h^{(r-1)} = (\hat{\mathbf{X}}_1, \hat{\mathbf{X}}_2, ..., \hat{\mathbf{X}}_{T_{r-1}}) = (\mathbf{X}_1, \mathbf{X}_2, ..., \mathbf{X}_{T_{r-1}})\mathbf{E}' \in$ $\mathbb{R}^{N\times C_{r-1}\times T_{r-1}}$ 以动态调整输入，通过合并相关信息。

时空卷积

时空注意力模块使网络能够自动地相对更多地关注有价值的信息。经过注意力机制调整的输入被送入时空卷积模块，该模块的结构如图5所示。这里提出的时空卷积模块由空间维度上的图卷积组成，用于从邻域中捕获空间依赖关系，以及沿时间维度进行的卷积，用于从邻近时间中利用时间依赖关系。

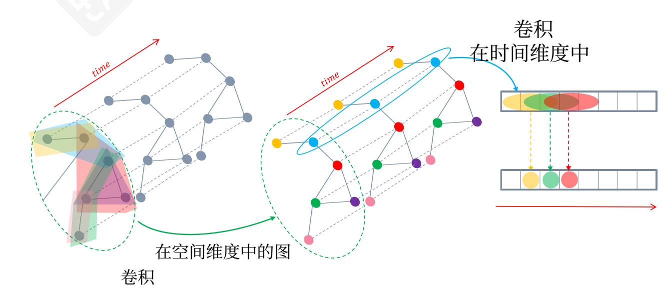


图5：ASTGCN的时空卷积架构

Graph convolution in spatial dimension The spectral graph theory generalizes the convolution operation from grid-based data to graph structure data. In this study, the traffic network is a graph structure in nature, and the features of each node can be regarded as the signals on the graph (Shuman et al. 2013). Hence, in order to make full use of the topological properties of the traffic network, at each time slice we adopt graph convolutions based on the spectral graph theory to directly process the signals, exploiting signal correlations on the traffic network in the spatial dimension. The spectral method transforms a graph into an algebraic form to analyze the topological attributes of graph, such as the connectivity in the graph structure.

In spectral graph analysis, a graph is represented by its corresponding Laplacian matrix. The properties of the graph structure can be obtained by analyzing Laplacian matrix and its eigenvalues. Laplacian matrix of a graph is defined as $\mathbf{L} = \mathbf{D} - \mathbf{A}$, and its normalized form is $\mathbf{L} = \mathbf{I}_N - \mathbf{D}^{-\frac{1}{2}}\mathbf{A}\mathbf{D}^{-\frac{1}{2}} \in \mathbb{R}^{N\times N}$, where $\mathbf{A}$ is the adjacent matrix, $\mathbf{I}_N$ is a unit matrix, and the degree matrix $\mathbf{D} \in \mathbb{R}^{N\times N}$ is a diagonal matrix, consisting of node degrees, $\mathbf{D}_{ii} = \sum_j \mathbf{A}_{ij}$. The eigenvalue decomposition of the Laplacian matrix is $\mathbf{L} = \mathbf{U}\mathbf{\Lambda}\mathbf{U}^T$, where $\mathbf{\Lambda} = diag([\lambda_0, ..., \lambda_{N-1}]) \in \mathbb{R}^{N\times N}$ is a diagonal matrix, and $\mathbf{U}$ is Fourier basis. Taking the traffic flow at time t as an example, the signal all over the graph is $x = \mathbf{x}_t^f \in \mathbb{R}^N$, and the graph Fourier transform of the signal is defined as $\hat{x} = \mathbf{U}^T x$. According to the properties of the Laplacian matrix, $\mathbf{U}$ is an orthogonal matrix, so the corresponding inverse Fourier transform is $x = \mathbf{U}\hat{x}$. The graph convolution is a convolution operation implemented by using linear operators that diagonalize in the Fourier domain to replace the classical convolution operator (Henaff, Bruna, and LeCun 2015). Based on this, the signal x on the graph G is filtered by a kernel g_θ:

$$g_\theta *_G x = g_\theta(\mathbf{L})x = g_\theta(\mathbf{U}\mathbf{\Lambda}\mathbf{U}^T)x = \mathbf{U}g_\theta(\mathbf{\Lambda})\mathbf{U}^T x \quad (5)$$

where $*_G$ denotes a graph convolution operation. Since the convolution operation of the graph signal is equal to the product of these signals which have been transformed into the spectral domain by graph Fourier transform (Simonovsky and Komodakis 2017), the above formula can be understood as Fourier transforming g_θ and x respectively into the spectral domain, then multiplying their transformed results, and doing the inverse Fourier transform to get the final result of the convolution operation. However, it is expensive to directly perform the eigenvalue decomposition on the Laplacian matrix when the scale of the graph is large. Therefore, Chebyshev polynomials are adopted in this paper to solve this problem approximately but efficiently (Simonovsky and Komodakis 2017):

$$g_\theta *_G x = g_\theta(\mathbf{L})x = \sum_{k=0}^{K-1} \theta_k T_k(\tilde{\mathbf{L}})x \quad (6)$$

where the parameter $\theta \in \mathbb{R}^K$ is a vector of polynomial coefficients. $\tilde{\mathbf{L}} = \frac{2}{\lambda_{max}}\mathbf{L} - \mathbf{I}_N$, λ_{max} is the maximum eigenvalue of the Laplacian matrix. The recursive definition of the Chebyshev polynomial is $T_k(x) = 2xT_{k-1}(x) - T_{k-2}(x)$, where $T_0(x) = 1$, $T_1(x) = x$. Using approximate expansion of Chebyshev polynomial to solve this formulation corresponds to extracting information of the surrounding 0 to $(K-1)^{th}$-order neighbors centered on each node in the graph by the convolution kernel g_θ. The graph convolution module uses the Rectified Linear Unit (ReLU) as the final activation function, i.e., ReLU$(g_\theta *_G x)$.

In order to dynamically adjust the correlations between nodes, for each term of Chebyshev polynomial, we accompany $T_k(\tilde{\mathbf{L}})$ with the spatial attention matrix $\mathbf{S}' \in \mathbb{R}^{N\times N}$, then obtain $T_k(\tilde{\mathbf{L}}) \odot \mathbf{S}'$, where $\odot$ is the Hadamard product. Therefore, the above graph convolution formula changes to $g_\theta *_G x = g_\theta(\mathbf{L})x = \sum_{k=0}^{K-1} \theta_k (T_k(\tilde{\mathbf{L}}) \odot \mathbf{S}')x$.

We can generalize this definition to the graph signal with multiple channels. For example, in the recent component, the input is $\hat{\boldsymbol{\mathcal{X}}}_h^{(r-1)} = (\hat{\mathbf{X}}_1, \hat{\mathbf{X}}_2, ..., \hat{\mathbf{X}}_{T_{r-1}}) \in \mathbb{R}^{N\times C_{r-1}\times T_{r-1}}$, where the feature of each node has C_{r-1} channels. For each time slice t, performing C_r filters on the graph $\hat{\mathbf{X}}_t$, we get $g_\theta *_G \hat{\mathbf{X}}_t$, where $\Theta = (\Theta_1, \Theta_2, ..., \Theta_{C_r}) \in \mathbb{R}^{K\times C_{r-1}\times C_r}$ is the convolution kernel parameter (Kipf and Welling 2017). Therefore, each node is updated by the information of the 0∼K-1 neighbors of the node.

Convolution in temporal dimension After the graph convolution operations having captured neighboring information for each node on the graph in the spatial dimension, a standard convolution layer in the temporal dimension is further stacked to update the signal of a node by merging the information at the neighboring time slice, as shown by the right part in Fig. 5. Also take the operation on the r^{th} layer in the recent component as an example:

$$\boldsymbol{\mathcal{X}}_h^{(r)} = ReLU(\Phi * (ReLU(g_\theta *_G \hat{\boldsymbol{\mathcal{X}}}_h^{(r-1)}))) \in \mathbb{R}^{C_r\times N\times T_r} \quad (7)$$

where $*$ denotes a standard convolution operation, Φ is the parameters of the temporal dimension convolution kernel, and the activation function is ReLU.

In conclusion, a spatial-temporal convolution module is able to well capture the temporal and spatial features of traffic data. A spatial-temporal attention module and a spatial-temporal convolution module forms a spatial-temporal block. Multiple spatial-temporal blocks are stacked to further extract larger range of dynamic spatial-temporal correlations. Finally, a fully connected layer is appended to make sure the output of each component has the same dimension and shape with the forecasting target. The final fully connected layer uses ReLU as the activation function.

Multi-Component Fusion

In this section, we will discuss how to integrate the outputs of the three components. Take forecasting the traffic flow on the whole traffic network at 8:00 am on Friday as an example. It can be observed that the traffic flows at some areas have obvious peak periods in the morning, so the outputs of the daily-period and weekly-period components are more crucial. However, there are no distinct traffic cycle patterns in some other places, thus the daily-period and weekly-period components may be helpless. Consequently, when

空间维度上的图卷积 图谱理论将卷积操作从基于网格的数据推广到图结构数据。在本研究中，交通网络本质上是一种图结构，每个节点的特征可以被视为图上的信号（Shuman等人，2013年）。因此，为了充分利用交通网络拓扑特性，在每个时间切片上，我们采用基于图谱理论的图卷积来直接处理信号，利用交通网络在空间维度上的信号相关性。谱方法将图转换为代数形式，以分析图的结构属性，如图结构中的连通性。

在谱图分析中，图通过其对应的拉普拉斯矩阵表示。通过分析拉普拉斯矩阵及其特征值，可以获得图结构的性质。图的拉普拉斯矩阵定义为 $\mathbf{L} = \mathbf{D} - \mathbf{A}$，其归一化形式为 $\mathbf{L} = \mathbf{I}_N - \mathbf{D}^{-\frac{1}{2}}\mathbf{A}\mathbf{D}^{-\frac{1}{2}} \in \mathbb{R}^{N\times N}$，其中A是邻接矩阵，$\mathrm{I}_N$是单位矩阵，度矩阵D $\in \mathbb{R}^{N\times N}$ 是对角矩阵，由节点度数 $\mathbf{D}_{ii} = \sum_j \mathbf{A}_{ij}$组成。拉普拉斯矩阵的特征值分解是$\mathbf{L} = \mathbf{U}\mathbf{\Lambda}\mathbf{U}^T$，其中 $\mathbf{\Lambda} = diag([\lambda_0, ..., \lambda_{N-1}]) \in \mathbb{R}^{N\times N}$是对角矩阵，U是傅里叶基。以时间 t 的交通流为例，图上的信号是 $x = \mathbf{x}_t^f \in \mathbb{R}^N$，信号的图傅里叶变换定义为 $\hat{x} = \mathbf{U}^T x$。根据拉普拉斯矩阵的性质，U是正交矩阵，因此相应的逆傅里叶变换是 $x = \mathbf{U}\hat{x}$。图卷积是通过使用在傅里叶域中对角化的线性算子来代替经典卷积算子实现的卷积运算（Henaff、Bruna和LeCun 2015）。基于此，图G 上的信号 x 通过核 g_θ进行滤波：

$$g_\theta *_G x = g_\theta(\mathbf{L})x = g_\theta(\mathbf{U}\mathbf{\Lambda}\mathbf{U}^T)x = \mathbf{U}g_\theta(\mathbf{\Lambda})\mathbf{U}^T x \quad (5)$$

$*_G$ 表示图卷积操作。由于图信号的卷积操作等于通过图傅里叶变换（Si- monovsky and Komodakis 2017）将信号转换到频域后的乘积，因此上述公式可以理解为分别将 g_θ 和 x 转换到频域，然后将它们的转换结果相乘，再进行逆傅里叶变换以得到卷积操作的最终结果。然而，当图规模较大时，直接对拉普拉斯矩阵进行特征值分解非常耗时。因此，本文采用切比雪夫多项式来近似但高效地解决这个问题（Si- monovsky and Komodakis 2017）：

$$K-1 g_\theta *_G x = g_\theta(\mathbf{L})x = \sum \theta_k T_k(\tilde{\mathbf{L}})x$$

(6)$k=0$

参数 $\theta \in \mathbb{R}^K$ 是多项式系数的向量。 $\tilde{}$L $= \frac{2}{\lambda_{max}}$ L $-$ I_N， λ_{max} 是拉普拉斯矩阵的最大特征值。切比雪夫多项式的递归定义是 $T_k(x) = 2xT_{k-1}(x) - T_{k-2}(x)$，其中 $T_0(x) = 1$ $T_1(x) = x$，。使用切比雪夫多项式的近似展开式来求解此公式对应于通过卷积核 g_θ 提取图中每个节点周围 0 到 $(K-1)^{th}$ 阶邻居的信息。图卷积模块使用修正线性单元（ReLU）作为最终激活函数，即 ReLU$(g_\theta *_G x)$。

为了动态调整节点之间的相关性，对于切比雪夫多项式的每一项，我们伴随 $T_k(\tilde{\mathbf{L}})$ 与空间注意力矩阵 $\mathbf{S}' \in \mathbb{R}^{N\times N}$，然后得到 $T_k(\tilde{\mathbf{L}}) \odot \mathbf{S}'$，其中 $\odot$ 是哈达玛积。因此，上述图卷积公式变为g $_\theta *_G x = g_\theta(\mathbf{L})x = \sum_{k=0}^{K-1} \theta_k (T_k(\tilde{\mathbf{L}}) \odot \mathbf{S}')x$。

我们可以将此定义推广到多通道的图信号。例如，在最近的组件中，输入是 $\hat{\boldsymbol{\mathcal{X}}}_h^{(r-1)} = (\hat{\mathbf{X}}_1, \hat{\mathbf{X}}_2, ..., \hat{\mathbf{X}}_{T_{r-1}}) \in \mathbb{R}^{N\times C_{r-1}\times T_{r-1}}$，其中每个节点的特征有 C_{r-1} 个通道。对于每个时间切片 t，在图 X_t 上执行 C_r 个滤波器，我们得到 $g_\theta *_G \mathrm{X}_t$，其中 $\Theta = (\Theta_1, \Theta_2, ..., \Theta_{C_r}) \in \mathbb{R}^{K\times C_{r-1}\times C_r}$ 是卷积核参数（Kipf 和 Welling 2017）。因此，每个节点通过 0∼K-1 个邻居节点的信息进行更新。

时间维度上的卷积 在空间维度上捕获了图中每个节点的邻近信息后，标准的时间维度卷积层被进一步堆叠，以通过合并邻近时间切片的信息来更新节点的信号，如图 5 的右半部分所示。同样，以最近的组件中的 r^{th} 层操作作为例：

$$\boldsymbol{\mathcal{X}}_h^{(r)} = ReLU(\Phi * (ReLU(g_\theta *_G \hat{\boldsymbol{\mathcal{X}}}_h^{(r-1)}))) \in \mathbb{R}^{C_r\times N\times T_r} (7)$$

其中， $*$ 表示标准卷积操作， Φ 是时间维度卷积核的参数，激活函数为ReLU。

总之，时空卷积模块能够很好地捕捉交通数据的时间和空间特征。时空注意力模块和时空卷积模块构成一个时空块。多个时空块堆叠起来，以进一步提取更大范围的动态时空相关性。最后，添加一个全连接层，以确保每个组件的输出与预测目标具有相同的维度和形状。最终的全连接层使用ReLU作为激活函数。

多组件融合

在本节中，我们将讨论如何整合三个组件的输出。以周五早上8点预测整个交通网络的交通流量为例。可以观察到，某些区域的交通流量在早晨有明显的高峰期，因此日周期和周周期的输出更为关键。然而，在其他一些地方没有明显的交通周期模式，因此日周期和周周期的组件可能无能为力。因此，当

the outputs of different components are fused, the impacting weights of the three components for each node are different, and they should be learned from the historical data. So the final prediction result after the fusion is:

$$\hat{\mathbf{Y}} = \mathbf{W}_h \odot \hat{\mathbf{Y}}_h + \mathbf{W}_d \odot \hat{\mathbf{Y}}_d + \mathbf{W}_w \odot \hat{\mathbf{Y}}_w \quad (8)$$

where $\odot$ is the Hadamard product. $\mathbf{W}_h$, $\mathbf{W}_d$ and $\mathbf{W}_w$ are learning parameters, reflecting the influence degrees of the three temporal-dimensional components on the forecasting target.

Experiments

In order to evaluate the performance of our model, we carried out comparative experiments on two real-world highway traffic datasets.

Datasets

We validate our model on two highway traffic datasets PeMSD4 and PeMSD8 from California. The datasets are collected by the Caltrans Performance Measurement System (PeMS) (Chen et al. 2001) in real time every 30 seconds. The traffic data are aggregated into every 5-minute interval from the raw data. The system has more than 39,000 detectors deployed on the highway in the major metropolitan areas in California. Geographic information about the sensor stations are recorded in the datasets. There are three kinds of traffic measurements considered in our experiments, including total flow, average speed, and average occupancy.

PeMSD4 It refers to the traffic data in San Francisco Bay Area, containing 3848 detectors on 29 roads. The time span of this dataset is from January to February in 2018. We choose data on the first 50 days as the training set, and the remains as the test set.

PeMSD8 It is the traffic data in San Bernardino from July to August in 2016, which contains 1979 detectors on 8 roads. The data on the first 50 days are used as the training set and the data on the last 12 days are the test set.

Preprocessing

We remove some redundant detectors to ensure the distance between any adjacent detectors is longer than 3.5 miles. Finally, there are 307 detectors in the PeMSD4 and 170 detectors in the PeMSD8. The traffic data are aggregated every 5 minutes, so each detector contains 288 data points per day. The missing values are filled by the linear interpolation. In addition, the data are transformed by zero-mean normalization $x' = x - mean(x)$ to let the average be 0.

Settings

We implemented the ASTGCN model based on the MXNet[1] framework. According to Kipf and Welling (2017), we test the number of the terms of Chebyshev polynomial $K \in \{1, 2, 3\}$. As K becomes larger, the forecasting performance improves slightly. So does the kernel size in the temporal dimension. Considering the computing efficiency and the degree of improvement of the forecasting performance, we set $K = 3$ and the kernel size along the temporal dimension to 3. In our model, all the graph convolution layers use 64 convolution kernels. All the temporal convolution layers use 64 convolution kernels and the time span of the data is adjusted by controlling the step size of the temporal convolutions. For the lengths of the three segments, we set them as: $T_h = 24$, $T_d = 12$, $T_w = 24$. The size of the predicting window $T_p = 12$, that is to say, we aim at predicting the traffic flow over one hour in the future. In this paper, the mean square error (MSE) between the estimator and the ground truth are used as the loss function and minimized by backpropagation. During the training phase, the batch size is 64 and the learning rate is 0.0001. In addition, in order to verify the impact of the spatio-temporal attention mechanism proposed here, we also design a degraded version of ASTGCN, named Multi-Component Spatial-Temporal Graph Convolution Networks (MSTGCN), which gets rid of the spatial-temporal attention. The settings of MSTGCN are the same as those of ASTGCN, except no spatial-temporal attention.

[1]https://mxnet.apache.org/

Baselines

We compare our model with the following eight baselines:

- HA: Historical Average method. Here, we use the average value of the last 12 time slices to predict the next value.
- ARIMA (Williams and Hoel 2003): Auto-Regressive Integrated Moving Average method is a well-known time series analysis method for predicting the future values.
- VAR (Zivot and Wang 2006): Vector Auto-Regressive is a more advanced time series model, which can capture the pairwise relationships among all traffic flow series.
- LSTM (Hochreiter and Schmidhuber 1997): Long Short-Term Memory network, a special RNN model.
- GRU (Chung et al. 2014): Gated Recurrent Unit network, a special RNN model.
- STGCN (Li et al. 2018): A spatial-temporal graph convolution model based on the spatial method.
- GLU-STGCN (Yu, Yin, and Zhu 2018): A graph convolution network with a gating mechanism, which is specially designed for traffic forecasting.
- GeoMAN (Liang et al. 2018): A multi-level attention-based recurrent neural network model proposed for the geo-sensory time series prediction problem.

Root mean square error (RMSE) and mean absolute error (MAE) are used as the evaluation metrics.

Comparison and Result Analysis

We compare our models with the eight baseline methods on PeMSD4 and PeMSD8. Table 1 shows the average results of traffic flow prediction performance over the next one hour.

It can be seen from Table 1 that our ASTGCN achieves the best performance in both two datasets in terms of all evaluation metrics. We can observe that the prediction results of the traditional time series analysis methods are usually not ideal, demonstrating those methods' limited abilities

不同组件的输出被融合时，三个组件对每个节点的权重影响不同，这些权重应该从历史数据中学习。因此，融合后的最终预测结果为：

$$\hat{\mathbf{Y}} = \mathbf{W}_h \odot \hat{\mathbf{Y}}_h + \mathbf{W}_d \odot \hat{\mathbf{Y}}_d + \mathbf{W}_w \odot \hat{\mathbf{Y}}_w \quad (8)$$

其中 $\odot$ 表示 Hadamard 积。W_h、W_d 和 W_w 是学习参数，反映了三个时间维度分量对预测目标的影响程度。

实验

为了评估我们的模型性能，我们在两个真实世界的高速公路交通数据集上进行了对比实验。

数据集

我们在加州的两个高速公路交通数据集 PeMSD4 和 PeMSD8 上验证了我们的模型。这些数据集由加州交通部性能测量系统 (PeMS) (Chen 等人，2001) 每隔 30 秒实时收集。交通数据从原始数据中每 5 分钟聚合一次。该系统在加州主要都市区的高速公路上部署了超过 39,000 个检测器。传感器站点的地理信息记录在数据集中。我们的实验中考虑了三种交通测量指标，包括总流量、平均速度和平均占有率。

PeMSD4 它指的是旧金山湾区（San Francisco Bay Area）的交通数据，包含29条路上的3848个检测器。该数据集的时间跨度为2018年1月至2月。我们选择前50天的数据作为训练集，其余作为测试集。

PeMSD8 它是圣贝尼托（San Bernardino）从2016年7月至8月的交通数据，包含8条路上的1979个检测器。前50天的数据用于训练集，最后12天的数据用于测试集。

预处理

我们移除了一些冗余的检测器，以确保任意相邻检测器之间的距离大于3.5英里。最终，PeMSD4中有307个检测器，PeMSD8中有170个检测器。交通数据每5分钟聚合一次，因此每个检测器每天包含288个数据点。缺失值通过线性插值填充。此外，数据通过零均值标准化 $x' = x - mean(x)$ 进行转换，使平均值变为0。

设置

我们基于MXNet[1]框架实现了ASTGCN模型。根据Kipf和Welling（2017）的研究，我们测试了切比雪夫多项式的项数 $K \in \{1, 2, 3\}$。随着 K 的增大，预测性能略有提升。时间维度上的核大小也是如此。考虑到计算效率和预测性能的提升程度，我们设定 $K = 3$ 以及时间维度上的核大小为3。在我们的模型中，所有图卷积层都使用64个卷积核。所有时间卷积层也使用64个卷积核，并通过控制时间卷积的步长来调整数据的时间跨度。对于三个片段的长度，我们设定为：$T_h = 24$ $T_d = 12$ $T_w = 24$、$T_p = 12$、。预测窗口的大小 $T_p = 12$，也就是说，我们的目标是预测未来一小时的交通流量。在本文中，我们使用估计值与真实值之间的均方误差（MSE）作为损失函数，并通过反向传播进行最小化。在训练阶段，批大小为64，学习率为0.0001。此外，为了验证我们提出的时空注意力机制的影响，我们还设计了一个ASTGCN的退化版本，称为多分量时空图卷积网络（MSTGCN），它摒弃了时空注意力机制。MSTGCN的设置与ASTGCN相同，除了没有时空注意力机制。

[1]https://mxnet.apache.org/

基线

我们与以下八个基线模型进行比较：

- HA：历史平均法。这里，我们使用过去12个时间片（time slices）的平均值来预测下一个值。
- ARIMA (Williams和Hoel 2003)：自回归积分移动平均法是一种著名的时间序列分析方法，用于预测未来值。
- VAR (Zivot and Wang 2006)：向量自回归是一种更高级的时间序列模型，能够捕捉所有交通流量序列之间的两两关系。
- LSTM (Hochreiter and Schmidhuber 1997)：长短期记忆网络，一种特殊的循环神经网络模型。
- GRU (Chung et al. 2014)：门控循环单元网络，一种特殊的循环神经网络模型。
- STGCN (Li et al. 2018)：一种基于空间方法的空间-时间图卷积模型。
- GLU-STGCN(Yu, Yin, and Zhu 2018)：一种带有门控机制的图卷积网络，专门为交通预测而设计。
- GeoMAN (Liang 等人 2018)：一种为地理感知时间序列预测问题提出的多级注意力循环神经网络模型。

均方根误差 (RMSE) 和平均绝对误差 (MAE) 被用作评估指标。

对比与结果分析

我们在 PeMSD4 和 PeMSD8 上将我们的模型与八个基线方法进行比较。表1 展示了未来一小时交通流量预测性能的平均结果。

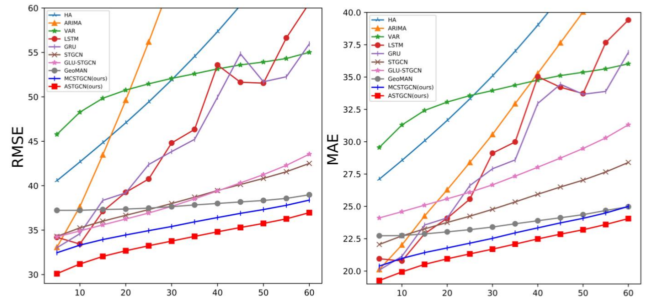
从表1可以看出，我们的ASTGCN在两个数据集上所有评估指标方面都取得了最佳性能。我们可以观察到，传统时间序列分析方法的预测结果通常不理想，这表明这些方法的局限性。

Model	PeMSD4		PeMSD8	
	RMSE	MAE	RMSE	MAE
HA	54.14	36.76	44.03	29.52
ARIMA	68.13	32.11	43.30	24.04
VAR	51.73	33.76	31.21	21.41
LSTM	45.82	29.45	36.96	23.18
GRU	45.11	28.65	35.95	22.20
STGCN	38.29	25.15	27.87	18.88
GLU-STGCN	38.41	27.28	30.78	20.99
GeoMAN	37.84	23.64	28.91	17.84
MSTGCN (ours)	**35.64**	**22.73**	**26.47**	**17.47**
ASTGCN (ours)	**32.82**	**21.80**	**25.27**	**16.63**

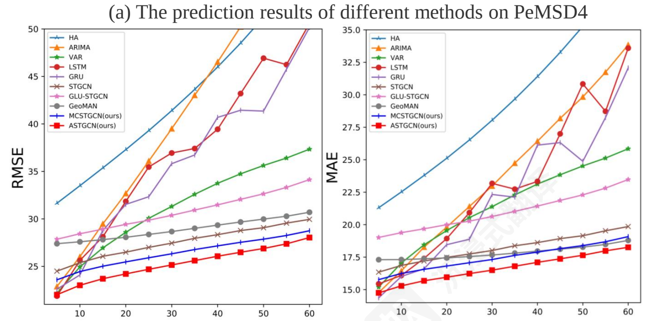
Table 1: Average performance comparison of different approaches on PeMSD4 and PeMSD8.

of modeling nonlinear and complex traffic data. By comparison, the methods based on deep learning generally obtain better prediction results than the traditional time series analysis methods. Among them, the models which simultaneously take both the temporal and spatial correlations into account, including STGCN, GLU-STGCN, GeoMAN and two versions of our model, are superior to the traditional deep learning models such as LSTM and GRU. Besides, GeoMAN performs better than STGCN and GLU-STGCN, indicating the multi-level attention mechanisms applied in GeoMAN are efficient in capturing the dynamic changings of traffic data. Our MSTGCN, without any attention mechanisms, achieve better results than the previous state-of-the-art models, proving the advantages of our model in describing spatial-temporal features of the highway traffic data. Then combined with the spatial-temporal attention mechanisms, our ASTGCN further reduces the forecasting errors.

Fig. 6 shows the changes of prediction performance of various methods as the prediction interval increases. Overall, as the prediction interval becomes longer, the corresponding difficulty of prediction is getting greater, hence the prediction errors also increase. As can be seen from the figure, the methods only taking the temporal correlation into account can achieve good results in the short-term prediction, such as HA, ARIMA, LSTM and GRU. However, with the increase of the prediction interval, their prediction accuracy drops dramatically. By comparison, the performance of VAR drops slower than those methods. This is mainly because VAR can simultaneously consider the spatial-temporal correlations which are more important in the long-term prediction. However, when the scale of the traffic network becomes larger, i.e., there are more time series considered in the model, the prediction error of VAR increases, as shown in Fig.6, its performance on PeMSD4 is worse than that on PeMSD8. The errors of deep learning methods increase slowly with prediction interval increases, and their overall performance is good. Our ASTGCN model achieves the best prediction performance almost all the time. Especially in the long-term prediction, the differences between ASTGCN and other baselines are more significant, showing that the strategy of combining attention mechanism with graph convolution can better mine the dynamic spatial-temporal patterns of traffic data.



(a) The prediction results of different methods on PeMSD4



(b) The prediction results of different methods on PeMSD8

Figure 6: Performance changes of different methods as the forecasting interval increases.

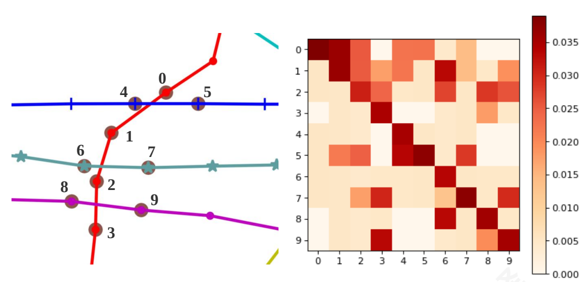


Figure 7: The attention matrix obtained from the spatial attention mechanism.

In order to investigate the role of attention mechanisms in our model intuitively, we perform a case study: picking out a sub-graph with 10 detectors from the PeMSD8 and showing the average spatial attention matrix among detectors in the training set. As shown on the right side of Fig. 7, in the spatial attention matrix, the i-th row represents the correlation strength between each detector and the i-th detector. For instance, look at the last row, we can know traffic flows on the 9th detector is closely related to those on the 3th and 8th detectors. This is reasonable since these three detectors are close in space on the real traffic network, as shown on the left side of Fig. 7. Hence, our model not only achieves a best forecasting performance but also shows an interpretability advantage.

模型	PeMSD4		PeMSD8	
	RMSE	MAE	RMSE	MAE
HA	54.14	36.76	44.03	29.52
ARIMA	68.13	32.11	43.30	24.04
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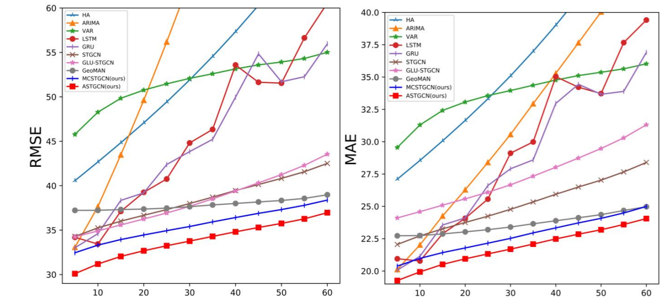
GeoMAN	37.84	23.64	28.91	17.84
MSTGCN (ours)	**35.64**	**22.73**	**26.47**	**17.47**
ASTGCN (ours)	**32.82**	**21.80**	**25.27**	**16.63**

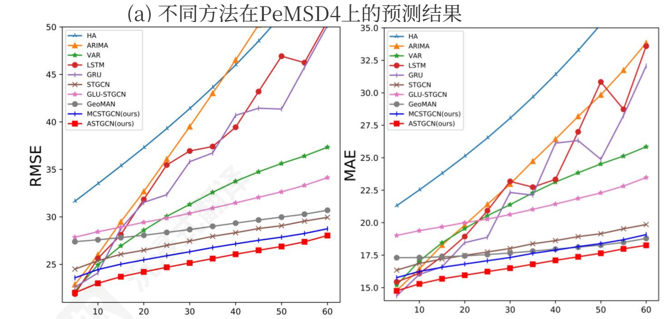
表1：不同方法在PeMSD4和PeMSD8上的平均性能比较。

从表1可以看出，我们的ASTGCN在两个数据集上所有评估指标方面均取得了最佳性能。我们可以观察到，传统时间序列分析方法的预测结果通常不理想，这表明这些方法的建模非线性复杂交通数据的能力有限。相比之下，基于深度学习的方法通常比传统时间序列分析方法获得更好的预测结果。其中，同时考虑时序和空间相关性的模型，包括STGCN、GLU-STGCN、GeoMAN以及我们模型的两个版本，均优于传统的深度学习模型如LSTM和GRU。此外，GeoMAN的性能优于STGCN和GLU-STGCN，表明GeoMAN中应用的多级注意力机制在捕捉交通数据的动态变化方面是有效的。我们的MSTGCN没有注意力机制，却取得了比以往最先进模型更好的结果，证明了我们的模型在描述高速公路交通数据时空特征方面的优势。然后结合时空注意力机制，我们的ASTGCN进一步降低了预测误差。

图6显示了各种方法的预测性能随预测区间的增加而变化的情况。总体而言，随着预测区间的变长，相应的预测难度也在增大，因此预测误差也随之增加。从图中可以看出，仅考虑时间相关性的方法在短期预测中可以取得良好结果，例如HA、ARIMA、LSTM和GRU。然而，随着预测区间的增加，它们的预测精度会急剧下降。相比之下，VAR的性能下降速度比这些方法要慢。这主要是因为VAR可以同时考虑时空相关性，而时空相关性在长期预测中更为重要。但是，当交通网络的规模变大时，即模型中考虑的时间序列更多时，VAR的预测误差会增加，如图6所示，其在PeMSD4上的性能比在PeMSD8上的性能要差。深度学习方法的误差随预测区间的增加而缓慢增加，其整体性能良好。我们的ASTGCN模型在大多数情况下都实现了最佳的预测性能。特别是在长期预测中，ASTGCN与其他基线的差异更为显著，表明结合注意力机制与图卷积的策略可以更好地挖掘交通数据的动态时空模式。



(a) 不同方法在PeMSD4上的预测结果



(b) 不同方法在PeMSD8上的预测结果

图6：随着预测间隔的增加，不同方法的性能变化。

结合注意力机制与图卷积的策略可以更好地挖掘交通数据的动态时空模式。

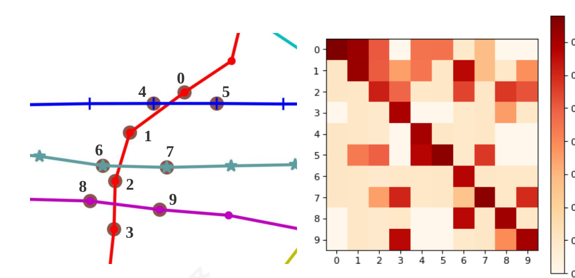


图7：从空间注意力机制获得的注意力矩阵。

为了直观地研究注意力机制在我们模型中的作用，我们进行了一个案例研究：从PeMSD8中挑选出包含10个检测器的子图，并展示训练集中检测器之间的平均空间注意力矩阵。如图7右侧所示，在空间注意力矩阵中，第 i行表示每个检测器与第 i个检测器之间的相关性强度。例如，查看最后一行，我们可以知道第9个检测器的交通流量与第3个和第8个检测器的交通流量密切相关。这是合理的，因为这三个检测器在真实交通网络中空间位置相近，如图7左侧所示。因此，我们的模型不仅实现了最佳预测性能，还展现了解释性优势。

Conclusion and Future Work

In this paper, a novel attention based spatial-temporal graph convolution model called ASTGCN is proposed and successfully applied to forecasting traffic flow. The model combines the spatial-temporal attention mechanism and the spatial-temporal convolution, including graph convolutions in the spatial dimension and standard convolutions in the temporal dimension, to simultaneously capture the dynamic spatial-temporal characteristics of traffic data. Experiments on two real-world datasets show that the forecasting accuracy of the proposed model is superior to existing models. The code has been released at: `https://github.com/wanhuaiyu/ASTGCN`.

Actually, the highway traffic flow is affected by many external factors, like weather and social events. In the future, we will take some external influencing factors into account to further improve the forecasting accuracy. Since the ASTGCN is a general spatial-temporal forecasting framework for the graph structure data, we can also apply it to other pragmatic applications, such as estimating time of arrival.

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结论与未来工作

本文提出了一种名为ASTGCN的新型基于注意力的时空图卷积模型，并将其成功应用于交通流量预测。该模型结合了时空注意力机制和时空卷积，包括空间维度上的图卷积和标准时间维度上的卷积，以同时捕捉交通数据的动态时空特征。在两个真实世界数据集上的实验表明，所提出的模型的预测精度优于现有模型。代码已发布在：https://github.com/wanhuaiyu/ASTGCN。

实际上，高速公路交通流受许多外部因素的影响，例如天气和社会事件。未来，我们将考虑一些外部影响因素，以进一步提高预测精度。由于 AST- GCN 是一种用于图结构数据的通用时空预测框架，我们也可以将其应用于其他实际应用，例如估计到达时间。

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